

QUICK QUIZ — Guess the AI!

"The engineer said that ____ will help."

1 · INTRODUCTION & MOTIVATION

- mBERT is used in real applications — hiring tools, translation, healthcare
- If biased, it may affect decisions differently per language
- We probe gender bias across 5 languages using fill-mask
- We test whether this bias reflects real-world corpus patterns

2 · RESEARCH SUB-QUESTIONS

- SQ1 — How does mBERT respond to fill-mask prompts across EN, NL, FR, DE, ES?
- SQ2 — Do grammatical-gender languages (FR/DE/ES) show different bias than non-grammatical (EN/NL)?
- SQ3 — Do mC4 corpus patterns align with mBERT bias scores?

4 · DATASET

mC4

1,000 docs/language · allenai/c4 (Hugging Face) · Python streaming
Used for SQ3: real-world corpus comparison

WinoBias

396 EN sentences · coreference-based gender bias benchmark
Used for validation: r = 0.794 agreement with templates

5 · METHODOLOGY

8 Templates → mBERT Fill-Mask → P(he)/P(she) → Bias Score

1 template × 8 occupations × 5 languages → fill-mask
→ P(he), P(she) per occupation → signed_bias score
→ aggregate per language → compare with mC4 corpus patterns

male_ratio = P(he) / (P(he) + P(she))

signed_bias = male_ratio − 0.5 (>0 = male-biased)

Template per language (8 occupations each):

EN: The {occ} said that [MASK] will help.

NL: De {occ} zei dat [MASK] zal helpen.

FR: Le* {occ} a dit que [MASK] va aider.

DE: Der* {occ} sagte, dass [MASK] helfen wird.

ES: El* {occ} dijo que [MASK] ayudara.

* Masculine article — potential confound in SQ2

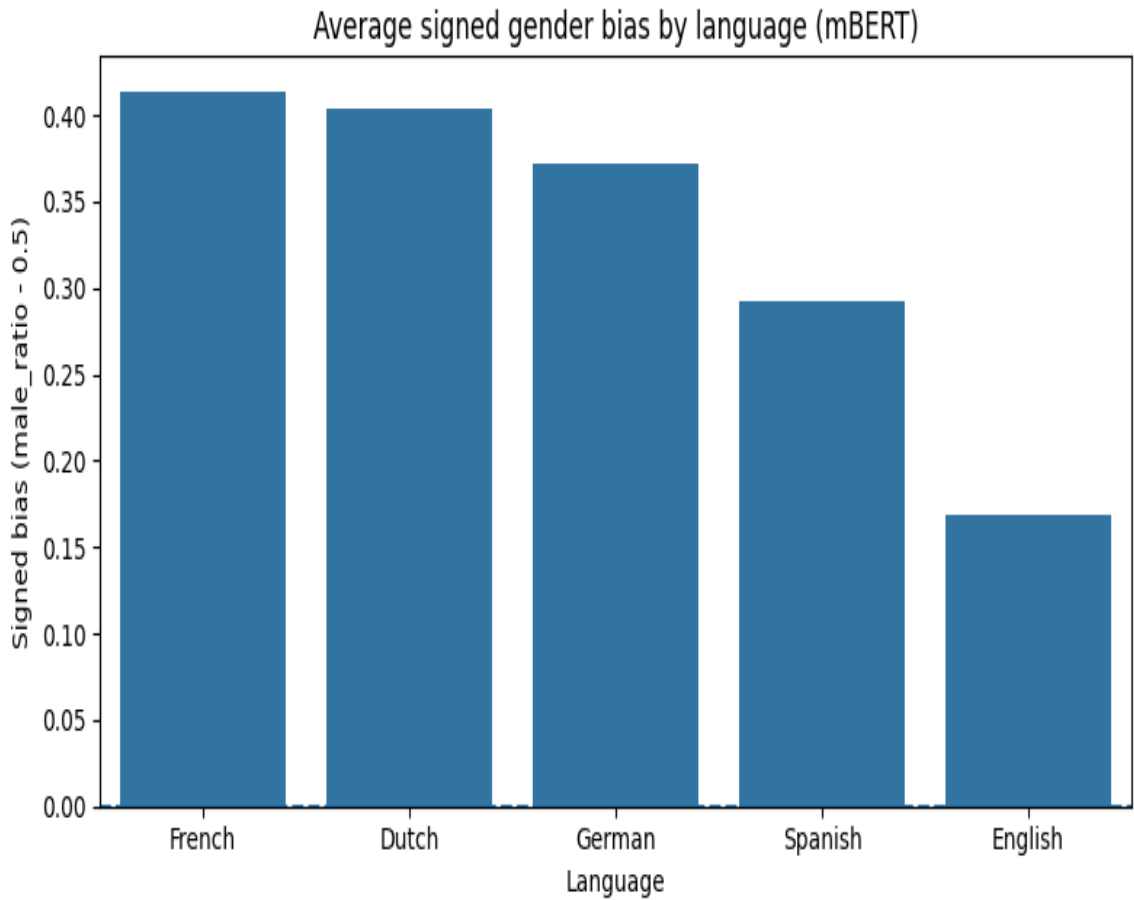
Occupations (8 per language):

Male-stereotyped: doctor · lawyer · engineer · manager

Female-stereotyped: nurse · teacher · assistant · secretary

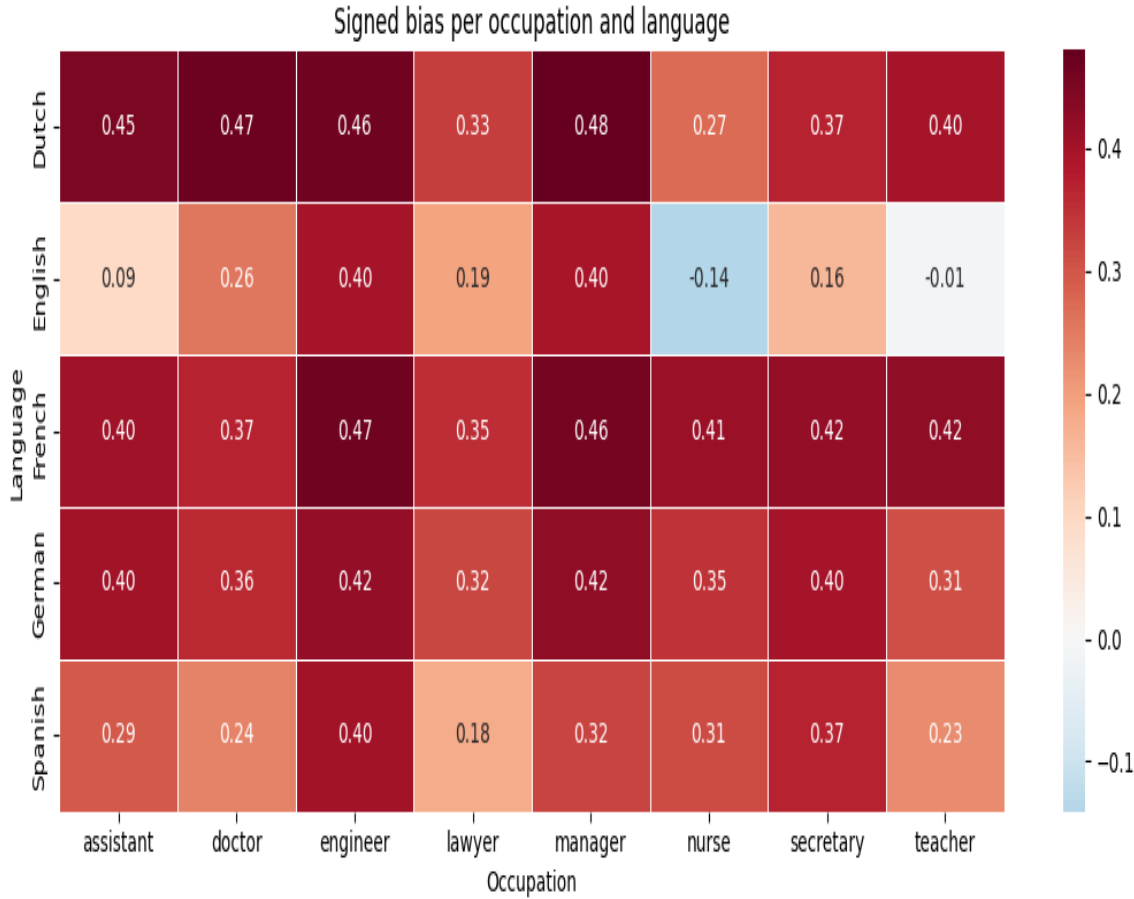
Validation vs WinoBias (EN): r = 0.794 — strong agreement

6 · RESULTS — SQ1: BIAS ACROSS ALL 5 LANGUAGES



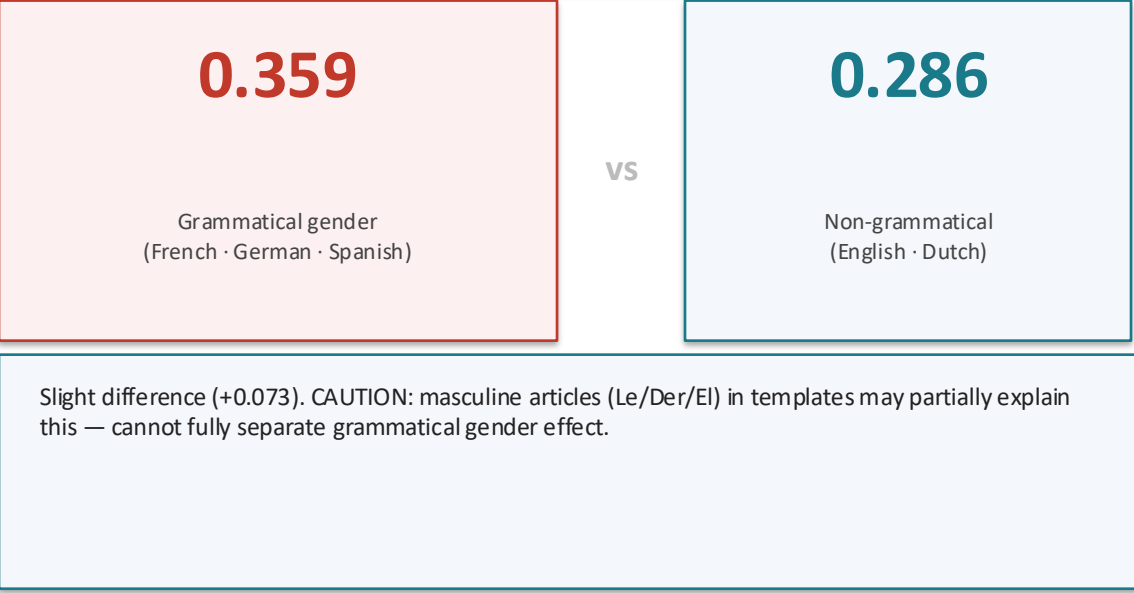
ALL 5 languages are male-biased. Order: FR (+0.413) > NL (+0.404) > DE (+0.372) > ES (+0.293) > EN (+0.169). English is the least biased.

7 · HEATMAP — BIAS PER OCCUPATION × LANGUAGE

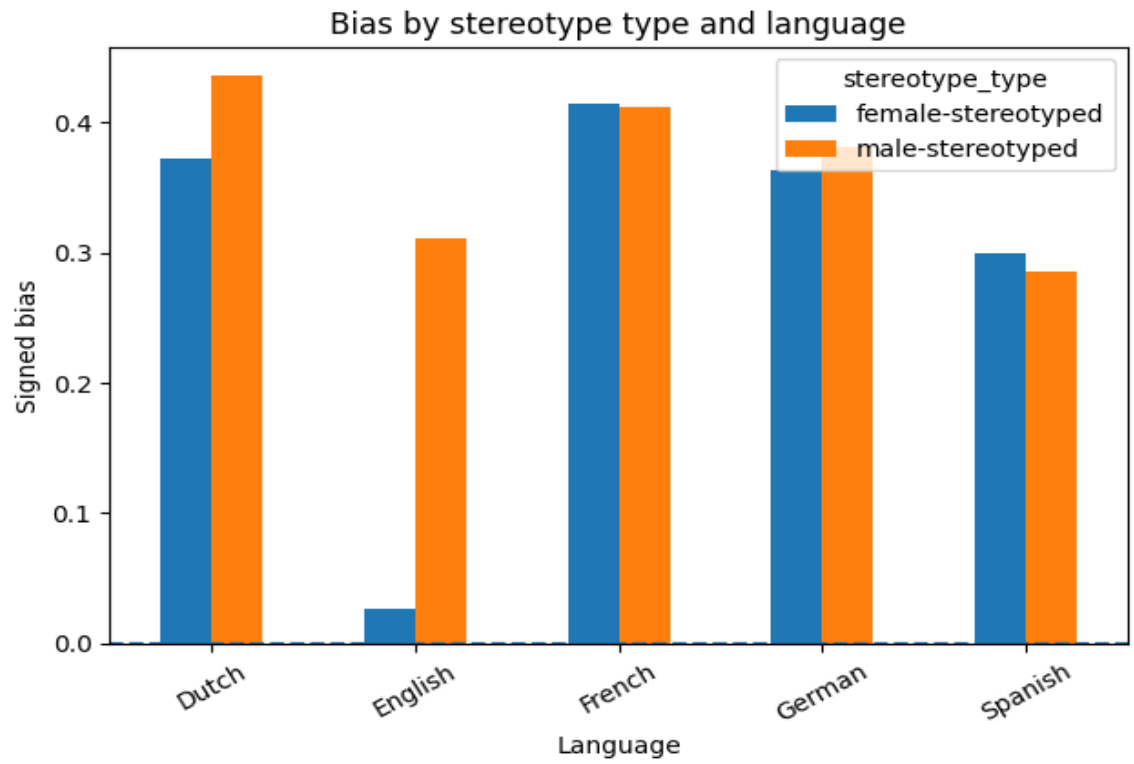


"Engineer" is the most biased occupation across ALL languages. Female-stereotyped occupations still show positive (male) bias.

8 · RESULTS — SQ2: GRAMMATICAL GENDER EFFECT



9 · BIAS BY OCCUPATION STEREOTYPE TYPE



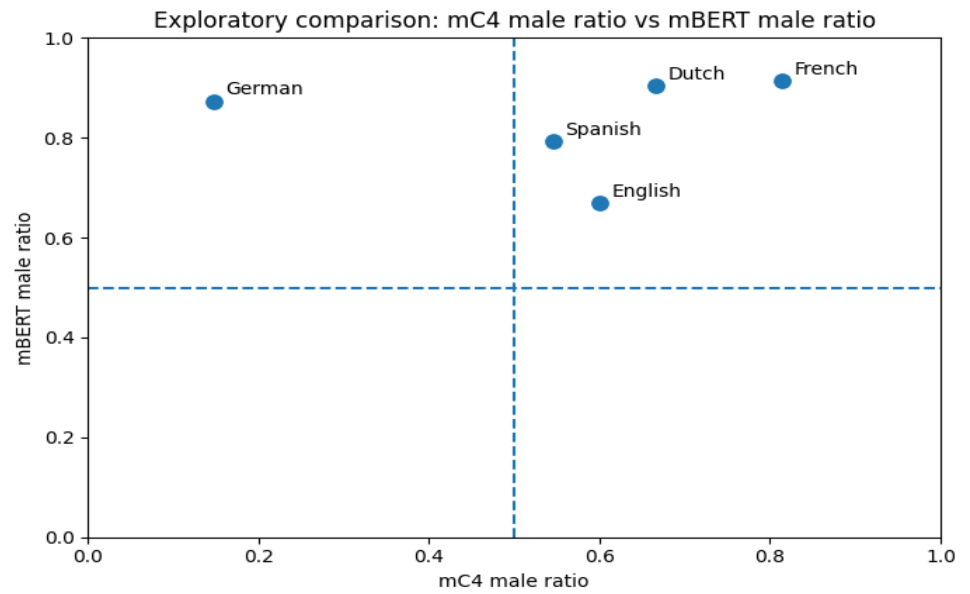
Male-stereotyped occupations show higher bias across all languages. Female-stereotyped still positive — no occupation is female-biased.

10 · SQ3: MC4 SIGNED BIAS VS MBERT SIGNED BIAS

Language	mBERT sig.bias	mC4 sig.bias	mC4 Sentences
French	+0.413	+0.315	116
Dutch	+0.404	+0.167	101
German	+0.372	-0.352 !!	94
Spanish	+0.293	+0.045	151
English	+0.169	+0.100	107

German: mBERT strongly male (+0.372) but mC4 is female-leaning (-0.352). Overall: r=0.057, p=0.927 — no significant correlation.

10B · SQ3 SCATTER: MC4 VS MBERT



r = 0.057

NO correlation

p = 0.927

Not significant

No correlation (r = 0.057, p = 0.927).
mBERT bias is not directly explained by real-world text.

11 · THE GERMAN OUTLIER — KEY FINDING

mC4: -0.352

Female-leaning in web text

≠

mBERT: +0.372

Strongly MALE biased

Why is German a complete outlier?

- ① Masculine grammatical forms (Der Arzt, Der Ingenieur)
- ② Cross-lingual transfer from English (dominant in training)
- ③ Wikipedia training differs from mC4 web text
- ④ Only ~27 German pronoun hits in mC4 — unreliable count

12 · LIMITATIONS

- ① One template per language — other structures may yield different results
- ② Small mC4 sample (1,000 docs) + keyword matching only → no coreference resolution
- ③ n=5 languages — Pearson r statistically underpowered
- ④ mC4 ≠ mBERT training data (Wikipedia) — approximate comparison
- ⑤ Masculine article confound in FR/DE/ES templates

13 · CONCLUSIONS & DISCUSSION

KEY FINDING

mBERT is male-biased in all 5 languages, with variation in strength (EN weakest, FR/NL strongest).
Corpus patterns do not explain model bias (r = 0.057).
This suggests that the bias is not directly explained by real-world text, but arises from training data and model behaviour.

THINK & DISCUSS

If you were building a multilingual AI:

- Would you try to remove the bias? How?
- Is any level of gender bias acceptable?

14 · KEY LITERATURE

Devlin et al. (2019)

BERT/mBERT architecture — multilingual pretraining on Wikipedia across 104 languages.

Kurita et al. (2019)

Fill-mask probing for gender bias in BERT — direct methodological basis for this study.